

The Non-Normal Grid — Master Organizer

Full narrative, classified claims, final theory, and results
Transient-aware dynamic analysis of inverter-dominated power systems

Jorge L. Mayorga Taborda Universidad de los Andes, Bogotá, Colombia

Reading key

■ **Green** = **verified**, computed and reproducible on the worked example. ■ **Orange** = **conjecture**, to be tested on the IEEE 39-bus system. The line between them is kept explicit on purpose: the paper must claim only the green and propose the orange.

Contents

1	The one-sentence thesis	1
2	The narrative, step by step	1
3	The final theory	2
4	Verified results (worked three-bus example)	2
5	Conjectures for the IEEE 39-bus study (3-month plan)	3
6	What is original, and against what	4
7	Implications	4
8	The single most powerful result	4

1 The one-sentence thesis

Core claim

Classical spectral hosting is the *graph-frequency-separable limit* of a more general picture. Heterogeneous damping and grid-following control make the grid operator non-normal, so a network can be stable and well-damped by every eigenvalue criterion yet amplify a disturbance several-fold before it decays. The resolvent — not the spectrum — is the object that sees this, and from it follow directed weak buses, a damping-placement rule, and a reinforcement rule that all diverge from their classical counterparts.

2 The narrative, step by step

Step 1 — The grid is a physical graph. From $I = Y_{\text{bus}}V$ and $S_i = V_i I_i^*$, the linearized active power gives a Jacobian equal to the weighted Laplacian, so the stacked dynamics are $M\ddot{\theta} + D\dot{\theta} + L\theta = p$. The graph is not an abstraction; it is the linearized power flow.

Step 2 — Two geometries, not one. The standard theory normalizes by mass alone, $\tilde{L} = M^{-1/2}LM^{-1/2}$, and certifies through $\nu_c = \lambda_2(\tilde{L})$. We keep damping as a second geometry, $\tilde{D} = M^{-1/2}DM^{-1/2}$, so the governing object is the quadratic pencil $(s^2I + s\tilde{D} + \tilde{L})v = 0$. Here $\tilde{L}$ locates soft (low-inertia) modes; $\tilde{D}$ locates poorly damped ones.

Step 3 — The hidden assumption. The scalar certificate is exact only when the modes separate, i.e. when $\tilde{L}$ and $\tilde{D}$ commute. Proportional damping commutes; inverter-synthesized damping does not, and the grid-following PLL adds a one-sided frequency-to-angle term. The operator A_{sys} becomes non-normal.

Step 4 — Why eigenvalues stop being enough. For a non-normal operator, $\|e^{A_{\text{sys}}t}\| \leq \kappa(V)e^{t\alpha}$ with $\kappa(V) \gg 1$. The conditioning $\kappa(V)$ is transient growth the eigenvalues do not bound. A normal stable system never amplifies ($\|e^{A_{\text{sys}}t}\| \leq 1$); a non-normal one can.

Step 5 — The resolvent diagnosis. The input–output filter is $H(\Omega) = C(i\Omega I - A_{\text{sys}})^{-1}B$, with peak $g_{\text{max}} = \sup_{\Omega} \sigma_{\text{max}}(H(\Omega))$. Its singular vectors at the peak separate the forcing bus from the response bus.

Step 6 — Design. The sensitivity $\partial g_{\text{max}}/\partial p$ for $p \in \{w_{ij}, M_i, D_i, \kappa_i\}$ turns the certificate into a design rule: which line to reinforce, where to add damping, where to convert to grid-forming.

3 The final theory

Definition 1 (Bi-metric operators and indices). $\tilde{L} = M^{-1/2}LM^{-1/2}$, $\tilde{D} = M^{-1/2}DM^{-1/2}$. Misalignment $\mathcal{C} = \left\|[\tilde{L}, \tilde{D}]\right\|_F$ (zero iff the geometries commute). Resolvent peak $g_{\text{max}} = \sup_{\Omega} \sigma_{\text{max}}(C(i\Omega I - A_{\text{sys}})^{-1}B)$. Directed weakness $W_{i \rightarrow j} = |[\psi_1]_i|^2 |[\phi_1]_j|^2$ from the singular vectors at the peak.

Theorem 1 (Classical limit)

If $[\tilde{L}, \tilde{D}] = 0$ and no asymmetric control term is present, the inertial graph-Fourier basis diagonalizes the response and the poles are $s_{k,\pm} = -\frac{\delta_k}{2} \pm \sqrt{(\frac{\delta_k}{2})^2 - \nu_k}$. Classical ν_c -based spectral hosting is exactly this separable limit.

Theorem 2 (Non-normal transient growth)

If $[\tilde{L}, \tilde{D}] \neq 0$ or the PLL term is present, A_{sys} is non-normal and $\sup_t \|e^{A_{\text{sys}}t}\| > 1$ is possible while $\alpha(A_{\text{sys}}) < 0$ and every modal damping ratio ζ_k exceeds the security threshold. The eigenvalues then do not bound the transient.

Theorem 3 (Certificate disagreement)

There exist reinforcements with $\partial \nu_c / \partial w_{ij} > 0$ (modal certificate improves) and $\partial g_{\text{max}} / \partial w_{ij} > 0$ (transient certificate worsens) simultaneously: the dynamic Braess effect, a sign disagreement between the two certificates.

4 Verified results (worked three-bus example)

N1 = synchronous, N2 = grid-following (PLL), N3 = grid-forming; $w = (5, 4, 3)$, $M = \text{diag}(10, 2, 6)$, $D = \text{diag}(0.4, 1.2, 0.5)$, $\kappa_p = 1$ unless noted.

V1 — Eigenvalue-safe but strongly non-normal

$\nu_c = 1.39 > 0$, all poles stable ($\max \text{Re} \lambda = -0.035$), yet $d_F(A_{\text{sys}}) = 4.12$ and resolvent peak $g_{\max} = 5.75$ at $\Omega^* = 1.16$ rad/s.

V2 — The decisive case: transient growth with healthy modes

A parameter search found a stable network with *every* modal damping ratio $\zeta \geq 0.108$ (well above the 0.05 security threshold), which the modal criterion calls safe, yet $\sup_t \|e^{A_{\text{sys}} t}\| = 3.94$: the system amplifies a disturbance almost fourfold before decaying. Since a normal stable system never amplifies, this growth is purely non-normal and invisible to the damping ratio. A random sweep found this in 483 of 483 stable modal-safe cases: the phenomenon is generic, not a corner case.

V3 — The second non-normality source is real

Stiffness–damping misalignment $\mathcal{C} = 1.22$ for inverter (heterogeneous) damping vs. $\mathcal{C} = 0$ for proportional damping; the off-diagonal of $\tilde{D}$ in the $\tilde{L}$ basis was 0.25 vs. 0. Heterogeneous damping couples the modes before any PLL is added.

V4 — Forcing bus $\neq$ response bus (directed weakness)

At the peak, optimal forcing ψ_1 concentrated on N3 (0.73), N1 (0.25); optimal response ϕ_1 on N1 (0.63), N3 (0.34). The most dangerous injection bus and the response bus differ — a directed weakness the short-circuit ratio cannot express.

V5 — Reinforcement and damping rankings diverge

Line reinforcement (+2 p.u.): L1–3 best by $\Delta \nu_c$ (+0.53) but L2–3 best by $\Delta g_{\max}$ (−0.78); L1–2 raises ν_c (+0.09) yet worsens the peak (+0.48) — a Braess case. Damping placement (+0.5): N3 helps most ($\Delta g_{\max} = -2.08$), the low-inertia GFL node N2 helps least (−0.29). Classical intuition picks the wrong target in both.

5 Conjectures for the IEEE 39-bus study (3-month plan)

C1 — Scaling

$d_F(A_{\text{sys}}(\rho))$, $\mathcal{C}$, and $g_{\max}$ rise with inverter penetration ρ , and inter-area modes make the transient-unsafe window wider than at three-bus scale.

C2 — Realistic-parameter decisive case

The transient-unsafe window (V2) persists with realistic PLL bandwidths and plausible penetration, not only the slow-PLL parameters where it was first found.

C3 — Ranking divergence at scale

The directed weak-bus ranking $W_{i \rightarrow j}$, the misalignment ranking from $\mathcal{C}$, and the damping-placement ranking diverge from the short-circuit-ratio and modal-participation rankings enough to change planning decisions.

C4 — Graph-frequency conversion as diagnosis

The off-diagonal of $H(\Omega)$ in the inertial graph-Fourier basis (“graph-frequency conversion”) is a useful localized diagnosis. *Caveat already found:* the proportional- damping case does not give exactly zero off-diagonal ($\mu_{\text{mix}} \approx 0.19$), so the clean “diagonal vs. converting” dichotomy must be stated carefully, not as a sharp law.

6 What is original, and against what

The honest novelty

The mathematics of non-classical damping (Caughey–O’Kelly 1965) and of resolvent/pseudospectral analysis (Trefethen–Embree) already exist. Spectral hosting (ν_c , Guo–Low, Dörfler) and modal analysis of IBR oscillations (Fan; coupling-mode benchmarks) already exist. The contribution is their *intersection*, absent from each alone: the joint stiffness–damping geometry $(\tilde{L}, \tilde{D})$ read through a directed-graph resolvent, applied to inverter hosting, yielding (i) a verified transient-unsafe regime with healthy modes, (ii) directed weak buses, (iii) damping- and reinforcement-placement rules that diverge from the classical ones.

Positioning, one line each. vs. spectral ν_c : it assumes proportional damping, so it cannot see $\mathcal{C}$. vs. complex-mode analysis: it ranks by damping ratio, which (V2) does not bound the non-normal transient. vs. Caughey: that framework is symmetric and passive; the PLL is asymmetric and active, outside it. vs. Grid-GSP: that processes measurements; here the filter is the dynamic operator itself.

7 Implications

Academic. A single bi-metric, non-normal resolvent picture recovers classical hosting as a limit and exposes a fragility (transient growth with healthy modes; directed pathways) that the scalar certificate cannot see by construction. It bridges power-system spectral analysis, structural-dynamics damping theory, and directed graph signal processing.

Industrial. A closed-form, simulation-free screen: one eigendecomposition and a frequency sweep. It tells a planner the bus to protect (response) is not the bus to keep clear of disturbances (forcing), flags reinforcements that would worsen the transient, and points where damping helps — a target that need not be the lowest-inertia node. It complements, not replaces, electromagnetic-transient studies, telling them where to look.

8 The single most powerful result

Headline (verified)

A grid can pass every eigenvalue and damping-ratio check — all modes stable, all $\zeta > 0.1$ — and still amplify a disturbance nearly fourfold before it decays, because the grid-following operator is non-normal. The damping ratio does not see it; the resolvent does. Verified as a generic phenomenon on the three-bus system; to be demonstrated at scale on the IEEE 39-bus.